

Sign Errors in Parabola Transformations

Answer Key

Algebra 1 Quadratic Functions

Quick Answers

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|-------|----------|-------|----------|
| 1. 2 | 3. -3, 5 | 5. 0 | 7. -3, 7 |
| 2. -3 | 4. 7 | 6. -5 | 8. — |
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Curriculum

Level: Algebra 1 Quadratic Functions

HSA-APR.B.3 – *problems 3, 7*

HSA-SSE.B.3 – *problems 1, 2, 4, 5, 6, 8*

Difficulty 1–5 of 5 across 8 tagged checks.

Common wrong answers

1. *If they got 102: read $x - 5$ as a vertex at $x = -5$ instead of $x = 5$*
 2. *If they got 61: wrote $(x - 4)$ squared, minus 3, for a shift LEFT 4, then evaluated at $x = -4$*
 4. *If they got -137: substituted $x = -6$, reading the sign inside the parentheses literally*
 5. *If they got 24: dropped the negative on the leading coefficient, so the reflection was lost*
 6. *If they got 191: read $x + 7$ as a shift RIGHT 7 and evaluated the minimum at $x = 7$*
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Grading note. On this sheet the sign reasoning is the skill, so a correct number reached by the wrong substitution should not earn full credit. The “common wrong answers” list above tells you which misreading produced each wrong value.

1. Minimum value of $f(x) = (x - 5)^2 + 2$.

The vertex x is whatever makes the squared factor zero: $x - 5 = 0$ gives $x = 5$, so the vertex is at $x = 5$, *not* $x = -5$. (The graph moved right, in the opposite direction from the printed sign.)

$$f(5) = (5 - 5)^2 + 2 = 0 + 2$$

Because $(x - 5)^2 \geq 0$ for every x , adding 2 can never produce anything smaller, so this really is the minimum. 2

If they got 102: they substituted $x = -5$, reading the printed sign literally.

2. $y = x^2$ shifted 4 left and 3 down; evaluate at $x = -4$.

A shift left is written with a *plus* inside the parentheses, because the graph moves to where $x + 4 = 0$, i.e. $x = -4$:

$$y = (x + 4)^2 - 3$$

$$y(-4) = (-4 + 4)^2 - 3 = 0 - 3$$

The value at $x = -4$ is the vertex height, which is the whole point of the check. -3
 If they got 61: they wrote $(x - 4)^2 - 3$ for “left 4” and then evaluated it at $x = -4$, so the two sign choices compounded.

3. x -intercepts of $p(x) = (x - 1)^2 - 16$.

Set $p(x) = 0$ and undo the transformations in reverse order:

$$(x - 1)^2 - 16 = 0$$

$$(x - 1)^2 = 16$$

$$x - 1 = \pm 4 \quad (\text{both roots — dropping } -4 \text{ loses an intercept})$$

$$x = 1 + 4 = 5 \quad \text{or} \quad x = 1 - 4 = -3$$

Sign check: the axis of symmetry is $x = 1$, and the two intercepts sit 4 units on either side of it ✓

$x = -3 \quad \text{and} \quad x = 5$

4. Find and fix: maximum of $y = -(x - 6)^2 + 7$.

The error: Jamal read $(x - 6)$ as a vertex at $x = -6$. The vertex sits where the squared factor is zero, so $x - 6 = 0$ gives $x = 6$. His arithmetic was fine; his h was not.

Correct substitution:

$$y(6) = -(6 - 6)^2 + 7 = -0 + 7$$

Since $a = -1 < 0$ the parabola opens downward, so this vertex value is a *maximum* — 7 is the largest output the function ever reaches. 7

If they got -137: they repeated Jamal’s substitution $x = -6$ instead of fixing it.

5. Reflect, stretch by 3, left 2, up 12; evaluate at $x = 0$.

Build the equation one transformation at a time. Reflection and vertical stretch both live in a : $a = -3$. Left 2 puts +2 inside the parentheses; up 12 adds 12 outside.

$$q(x) = -3(x + 2)^2 + 12$$

$$q(0) = -3(0 + 2)^2 + 12 = -3(4) + 12 = -12 + 12$$

So $x = 0$ is one of the x -intercepts of the transformed parabola. 0

If they got 24: they kept $a = +3$ and lost the reflection. Note the vertex is at $x = -2$ with value 12, so a positive a would make 12 a minimum instead of a maximum — the shape of the answer changes, not just the number.

6. Find and fix: minimum of $g(x) = (x + 7)^2 - 5$.

The error: Mia called $(x + 7)$ a shift 7 units *right*. It is a shift 7 units *left*, because $x + 7 = 0$ at $x = -7$. Her “down 5” was correct.

Because she put the vertex at $x = 7$, she evaluated at the wrong place. Correct substitution:

$$g(-7) = (-7 + 7)^2 - 5 = 0 - 5$$

-5

If they got 191: that is $g(7)$ — Mia’s value. It is a real point on the parabola, just not the lowest one, which is exactly why a wrong h can go unnoticed.

7. Reflect, right 2, up 25; find the x -intercepts.

Reflection makes $a = -1$; right 2 gives $(x - 2)$; up 25 adds 25:

$$y = -(x - 2)^2 + 25$$

Set it to zero:

$$-(x - 2)^2 + 25 = 0$$

$$(x - 2)^2 = 25$$

$$x - 2 = \pm 5$$

$$x = 7 \quad \text{or} \quad x = -3$$

$x = -3 \quad \text{and} \quad x = 7$

Why the reflection does not move the axis: reflecting over the x -axis replaces each output y with $-y$ and leaves every input untouched. The axis of symmetry is a statement about inputs, so it stays at $x = 2$ — and indeed -3 and 7 are each 5 units from 2.

8. Error analysis — Ravi’s study card. (open response — read the explanations)

R1 is wrong. The shift is not in the direction of the printed sign. The vertex lands where the squared factor is zero, so $(x + 4)$ puts the vertex at $x = -4$: the graph moves *left* 4. *Corrected:* “In $y = a(x - h)^2 + k$ the graph shifts to $x = h$, the value that makes $x - h$ zero. So $(x + 4)$ means $h = -4$: left 4.”

R2 is correct. k is added to every output after the squaring, so a positive k raises the whole graph and a negative k lowers it. Here the printed sign and the direction agree, because nothing is subtracted *from* y .

R3 is wrong. a multiplies the outputs; it does nothing to the inputs. A negative a flips the parabola upside down about its own vertex, but the vertex — and therefore both shifts — stays exactly where h and k put it. *Corrected:* “When $a < 0$ the parabola opens downward and its vertex becomes a maximum instead of a minimum; the shifts are unchanged.”

Full credit requires naming R1 and R3 as wrong, an explanation that appeals to “what makes the parentheses zero” (R1) and to “ a scales outputs, not inputs” (R3), and a corrected wording for each. Naming the wrong rules with no reason is partial credit.

R1 wrong, R2 correct, R3 wrong